

UNITE: Uncertainty-based Health Risk Prediction Leveraging Multi-sourced Data

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ABSTRACT

Successful health risk prediction demands **accuracy** and **reliability** of the model. Existing predictive models mainly depend on electronic health records (EHR). To further improve **model accuracy**, patients' geographical location, social economic status, environmental factors and detailed demographic information are all predictive signals that are missing from existing models. To achieve **model reliability**, the model needs to provide accurate prediction and uncertainty score of the prediction. However, existing uncertainty estimation approaches often failed in handling high-dimensional data, which are present in multi-source data.

To fill the gap, we propose **UNcertaInTy**-based hEalth risk prediction (UNITE) model. Building upon an adaptive multimodal deep kernel and a stochastic variational inference module, UNITE provides accurate disease risk prediction and uncertainty estimation leveraging multi-sourced health data including location, social economic and environmental factors. We evaluate UNITE on real-world disease risk detection tasks: nonalcoholic fatty liver disease (NASH) and Alzheimer's disease (AD). UNITE achieved up to 0.927 in Cohen's Kappa for AD detection, up to 0.609 in PR-AUC for NASH detection and outperforms various state-of-the-art baselines by up to 19% over the best baseline. We also show UNITE can model meaningful uncertainties and can provide evidence-based clinical support by clustering similar patients.

1 INTRODUCTION

With the collection of multi-sourced data including electronic health records (EHR) and other health related data sources such as location, environmental and social economic factors, health risk prediction from such multi-sourced data becomes a promising yet challenging task. From the application perspective, *existing works mainly consider EHR data as model inputs and ignore the importance of environmental and location risk factors* [1–3, 24, 25]. However, other data sources are also predictive and should be included in the model. In practice, many location-based health factors are important for predicting the health risks but are not recorded in EHR. For example, many studies have shown that environmental factors become important risk factors to the onset and progression for diseases like Alzheimer's disease (AD) [15] and nonalcoholic steatohepatitis (NASH) [18].

As multiple data sources are added to the model, input data become high-dimensional, complex and noisy, which often leads to high uncertainty in model prediction. Over the years, both deep learning (DL) models and probabilistic deep learning models (e.g., Bayesian neural networks and Gaussian process) have demonstrated early success in health risk predictive models.

Deep Learning. Traditional DL models often capture disease patterns based on patients' personal health trajectory [1–3, 24, 25] and output a probability score that indicates the likelihood of a patient who will develop a target disease. However, they are known to be prone to overconfidence and are difficult to produce effective patient-level uncertainty scores [21]. This affects the model reliability especially in the clinical context. Since even a model has a good average performance for a patient population, there still can be out-of-distribution (OOD) patients whose predictions are quite uncertain but often difficult to detect.

Probabilistic Deep Learning. To provide uncertainty estimation, the probabilistic deep learning including Bayesian neural networks (BNN) [31] and sparse Gaussian processes (GP) [23] has drawn recent interest. BNN models such as [32] learn a distribution over model parameter weights and use the uncertainty of the parameter weights to estimate the uncertainty of predictions. However, this approach usually requires a compromise between model complexity and expressiveness of variational distributions. Recently, there are also some attempts [11, 19] in modifying DNNs to approximate Bayesian, but they are not as flexible as non-parametric models such as GP. Compared with BNN models, the GP models such as [5, 8–10] are more expressive, but at the computational expense of the need to store and process the data points for the covariance. Thus, existing GP-based models are unable to model high-dimensional EHR data with reasonable efficiency and expressiveness.

Our Approach. To tackle all the aforementioned challenges, we propose UNITE, an uncertainty-based risk prediction model. Built upon an adaptive multimodal deep kernel and a stochastic variational inference module, UNITE provides accurate disease risk prediction and uncertainty estimation leveraging multi-sourced health data including environmental factors. UNITE is enabled by the following technical contributions.

- (1) **A stochastic variational inference module to provide accurate prediction and patient-level uncertainty score.**

To provide informative predictions based on global correlations, we formulate a population-level GP classification task. We introduce non-Gaussian likelihood, stochastic variational

inference (SVI), and inducing point method to enhance computational tractability. Moreover, UNITE is able to provide meaningful patient-level uncertainty scores, which is a key step in applying these models in clinical decision support applications.

- (2) **An adaptive deep kernel to integrate environmental and other location-based health risk factors.** The deep kernel module first applies to different data modalities, then uses a weight-based fusion network to learn the shared representation across all data modalities. The deep kernel module can effectively extract latent feature representations from multiple data sources.
- (3) **Comprehensive evaluation and benchmarking on health risk prediction** We compare with 8 baselines (4 DL and 4 probabilistic deep learning methods) and 2 variants of UNITE on two disease prediction tasks for Nonalcoholic fatty liver disease (NASH) and Alzheimer. We utilize two large real-world longitudinal patient datasets and their location, environmental and social economic factors in the experiment. Overall, UNITE significantly outperformed all the other baselines with .609 PRAUC for NASH and 0.927 in Cohen's Kappa for AD. We also show the learned uncertainty score could potentially be used to identify truly uncertain patients that are easily misdiagnosed and thus improve the overall prediction.

2 RELATED WORK

Deep learning models for health risk prediction mainly extract temporal patterns or graph structures from patients' historical diagnosis and medication records in their EHR data. For example, [1, 22] leveraged recurrent neural networks and its variant to better capture temporal health patterns. [2, 24] used bidirectional RNN and attention to capture the relationships of different visits for the prediction. [3] and [4] learned and exploited the graphical structure inherent in medical codes. In addition, [12, 20, 25] also considered patient demographics information in their models. However, in real world, there are other factors, such as location-based health factors, also affect the health risks.

In this work, we aim to utilize data from multiple sources and build UNITE as a multimodal learning framework.

Probabilistic deep learning for predictive health include Bayesian neural networks (BNN) [31] and sparse Gaussian processes (GP) [23] has drawn recent interest. For example, [32] proposed to use Bayesian neural networks (BNN) to capture model uncertainty. It learns a distribution over model weights and pass the uncertainty of the weights to eventually estimate the uncertainty in the predictions. However, this approach usually requires a compromise between model complexity and expressiveness. Also it could not learn population-level correlation among patients.

Compared with BNN models, GP models are more flexible and expressive. Recent attempts include [8–10], which use multi-task GP models for individual sepsis detection. However, since GP models are expensive in storing data points for the covariance matrix, these works either focus on low dimensional numerical features or decompose the task into separate modeling without considering the global effect. Recently, [5] proposed to augment GP with a deep

mean function for learning the global trend and a deep kernel for individual variability. However, it could not learn population-level correlation among patients.

There are also attempts [7, 11, 19] in modifying DNNs to approximate Bayesian, but are not as flexible as non-parametric models such as GP.

In this work, we proposed a GP-based model that has the following benefits: (1) It inherits the expressive power of GP and is augmented by a multimodal deep kernel that fuses multi-sourced data for improved representation power. (2) It reduces the computation cost by leverage stochastic variational inference with inducing points [29, 30] as an approximate to handle large-scale input data. (3) The covariance matrix provides population-level patient correlations, which could benefit evidence-based medicine.

3 METHOD

3.1 Problem Formulation

Given longitudinal EHR $E := \{X_i, y_i\}_{i=1}^P$ of P patients, where patient i is represented by a multivariate sequence $X_i := \{x_1^{(i)}, x_2^{(i)} \dots x_{T_i}^{(i)}\}$.

Here $x_{T_i}^{(i)}$ represents a medical code indicating the diagnosis or medication that patient i receives at time T_i . The groundtruth targets are given as $y_i \in \{0, 1\}$ where 1 means the patient has the disease and 0 means not. In addition to the EHR dataset, we also have the access to the patient's demographics data $D := \{S_i\}_{i=1}^P$ of P patients. $S_i := \{s_1^{(i)}, s_2^{(i)}, \dots, s_N^{(i)}\}$ denotes the demographics of the patient i .

Moreover, we denote the location-based health risk data (open source on CDC website) as $G := \{G_k\}_{k=1}^L$ of L locations. $G_k := \{g_1^{(k)}, g_2^{(k)}, \dots, g_M^{(k)}\}$, where each $g_l^{(k)}$ denotes one of the location-based factors at location k .

PROBLEM 1 (UNCERTAINTY-BASED HEALTH RISK PREDICTION). *Given the data from multiple modalities E, D, G , our objective is to provide informative health risk prediction. Specifically, for a patient i at location k with EHR input $\{X_i\}$, demographics features $\{S_i\}$, and the location-based factors $\{G_k\}$, the model aims to output a health risk distribution $p(y)$, such that (1) by evaluating the mean of $p(y)$, i.e., $\mathbb{E}(y)$, we can access the predicted health risk that indicates the probability of having the target disease; and (2) by computing the variance of $p(y)$, i.e., $\text{var}(y)$, we can quantify the uncertainty estimations about the model predictions.*

Table 1: Notations used in UNITE

Symbol	Definition and description
E	The EHR dataset
X_i	The EHR record of patient i
x_T^i	The medical code of patient i at time T
D	The demographics dataset
$s_j^{(i)}$	The j – th demographics features of patient i
G	The location-based dataset
$g_l^{(k)}$	The l – th location-based features at location k
$y \in \{0, 1\}$	Patient's disease label

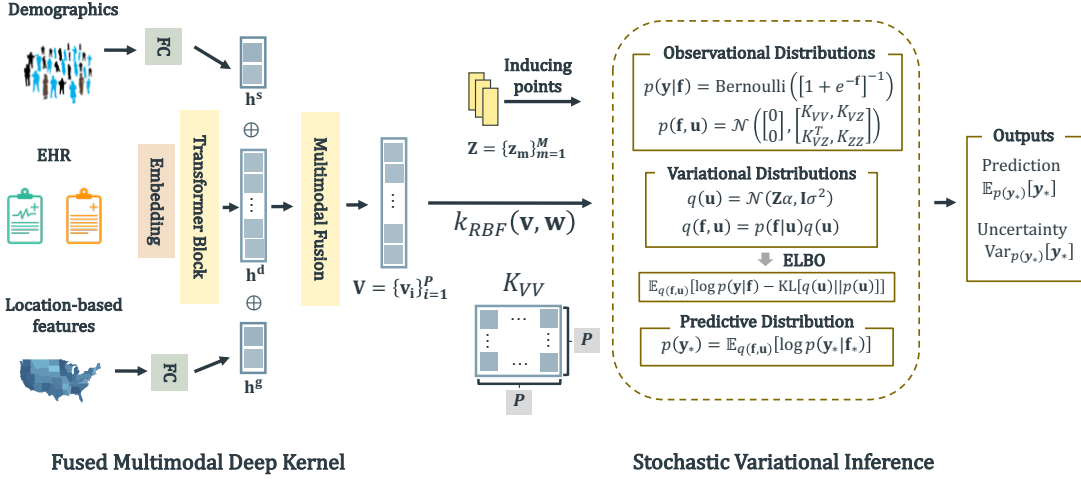


Figure 1: The UNITE model comprises of two components: the fused multimodal deep kernel module and stochastic variational inference module. (I) In fused multimodal deep kernel, we integrate data from multiple sources with different embedding mechanism, and then fuse them using a weight-based fusion technique to learn a shared embedding. After that, we warp the embedding networks with an radial basis function (RBF) kernel. (II) The embedding are then fed into a stochastic variational inference (SVI) module for making predictions as well as generating uncertainty scores based on global correlations. We also equip the SVI module with inducing points for efficient training.

3.2 The UNITE Model

UNITE comprises of two modules: the fused multimodal deep kernel module and the stochastic variational inference module. The fused multimodal deep kernel module first integrates data from multiple sources with different embedding mechanism, fuses them using a weight-based fusion technique to learn a shared embedding, and warps the embedding networks with a radial basis function (RBF) kernel. While stochastic variational inference (SVI) module takes the learned embedding to make predictions as well as perform uncertainty estimation based on global correlations. It is also equipped with an inducing point for efficient training.

(I) Fused Multimodal Deep Kernel Module

We leverage deep neural structures to effectively learn the meaningful feature representations from rich data. Specifically, we design different embedding frameworks for various data modalities. Then to learn a shared representation of comprehensive input data, we leverage an adaptive information fusion mechanism. The feature embedding networks are pre-trained using a linear classifier with cross-entropy loss to capture meaningful latent representations. The final step is to warp the latent vectors with a base RBF kernel.

EHR Records Embeddings. The EHR record of a patient includes a sequence of the diagnosis and medical codes for each visit. To learn the sequence embedding, we first tokenize the medical codes into sequences of integers (integers are indexes of tokens in a dictionary) as given by Eq. (1).

$$\mathbf{h}_0^d = \text{Tokenizer}(\mathbf{X}_i) \quad (1)$$

Then we utilize a multi-layer transformer to learn the latent representations. Transformer, which is a multi-head attention based

architecture, is able to learn the spatial relationships in the patient EHR records. In general, the attention mechanism maps a query and a set of key-value pairs to an output. The output is the weighted sum of the values, as the weight computed by a function of the corresponding query key pair. Since the self-attention mechanism contains no recurrence, it is a common practice that we add a position encoding to provide the sequential relationships of the input sequence as well [28].

We build multi-layer transformer blocks to capture the rich information in high-dimensional data. In detail, each transformer block sequentially applies a multi-head self attention layer, a normalization layer, a feed forward layer, and another normalization layer. We also add residual connections before the normalization to help the networks train faster. Specifically, given EHR record for patient i : $\mathbf{X}_i := [x_1^{(i)}, \dots, x_{T_i}^{(i)}]$, the latent embedding is given by Eq. (2).

$$\mathbf{h}_{n+1}^d = \text{Transformer}(\mathbf{h}_n^d) \quad (2)$$

where n is a hyperparameter indicating the number of transformer blocks applied.

Demographics and Location-based Features Embeddings. Furthermore, the demographics and location-based features from different locations are embedded into the same hidden space. For those numerical and categorical data, we leverage multi-layer perceptrons (MLP) to learn the latent representations as given by Eq. (3).

$$\mathbf{h}^s = \text{MLP}(\mathbf{s}); \quad \mathbf{h}^g = \text{MLP}(\mathbf{g}) \quad (3)$$

Adaptive Information Fusion. We demonstrate how to fuse the learned three latent embeddings $\mathbf{h}^d, \mathbf{h}^s, \mathbf{h}^g$ from different data modalities. To efficiently learn a shared embedding across them, we adopt a

parametric-matrix-based fusion technique as given by Eq. (4).

$$\mathbf{v} = \mathbf{h}^d \circ \mathbf{W}_d \oplus \mathbf{h}^s \circ \mathbf{W}_s \oplus \mathbf{h}^g \circ \mathbf{W}_g, \quad (4)$$

where $\circ$ is a Hadamard product, i.e., element-wise multiplication, and $\oplus$ is the vector concatenation operator. $\mathbf{W}_s, \mathbf{W}_d, \mathbf{W}_g$ are the learnable parameters that adjust the weights of the latent factors from different data sources.

Note that our proposed data fusion mechanism retains the flexibility to integrate any additional data modality by concatenating the modality embedding transformed by the corresponding weight matrix.

Pre-training Initialization. We pre-train the fused multimodal embedding networks for better initialization, following the implementation in [30]. We first apply a fully connected layer to the embedding outputs as a classifier. Then we pre-train the whole networks using Adam optimizer with the cross entropy loss objective in a supervised manner. In practice, by pre-trained initialization, we give the model a better prior to help the model converge fast.

RBF Kernel Warping. Using the individual level latent vectors obtained from Eq. (4), we can learn the population level covariance matrix among all the patients. Hence, we warp the fused latent vector outputs with a base kernel. In this work, we adopt the RBF kernel as it is widely-used in existing deep kernel method [29]. Given a pair of latent vectors $\mathbf{v}, \mathbf{w} \in \mathbf{V}$, the RBF kernel function is defined as in Eq. (5).

$$k_{RBF}(\mathbf{v}, \mathbf{w}|\gamma) = \exp\left(-\frac{1}{2} \sum_{k=1}^D \left(\frac{v_k - w_k}{l_k}\right)^2\right) \quad (5)$$

where $\mathbf{l} = \{l_k\}_{k=1}^D$ is the vector of length scale parameters and D is the size of latent vector. For the rest of the paper, we denote the kernel matrix computed on $\mathbf{V}$ as K_{VV} .

(II) Stochastic Variational Inference Module

Exact inference and learning in Gaussian process model with a non-Gaussian likelihood is not analytically tractable. Motivated by [29], we leverage stochastic variational inference with inducing points as an approximate technique to deal with large-scale input data.

Specifically, given the fused vector for patient i as $\mathbf{V} = \{\mathbf{v}_i\}_{i=1}^P$, we use Bernoulli likelihood to tackle the classification problem as in Eq. (6).

$$(\mathbf{y}|\mathbf{f}) \sim \text{Bernoulli}\left(\frac{1}{1 + e^{-\mathbf{f}}}\right); \quad \mathbf{f}(\mathbf{V}) \sim \mathcal{N}(\mathbf{0}, K_{VV}) \quad (6)$$

where $\mathbf{f}$ is generally known as the signal function that follows a Gaussian Process prior. K_{VV} is the covariance matrix (a.k.a. kernel) as defined in Eq. (5).

Let $\mathbf{Z} = \{\mathbf{z}_m\}_{m=1}^M$ be the collection of inducing points and $\mathbf{u}$ their corresponding signal. We then can express the joint signal distribution as given by Eq. (7).

$$(\mathbf{f}, \mathbf{u}) \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} K_{VV} & K_{VZ} \\ K_{VZ}^\top & K_{ZZ} \end{bmatrix}\right) \quad (7)$$

Therefore, the signal distribution $\mathbf{f}$ conditioned on the inducing points is given by Eq. (8).

$$(\mathbf{f}|\mathbf{u}) \sim \mathcal{N}(K_{VZ}K_{ZZ}^{-1}\mathbf{u}, K_{VV} - K_{VZ}K_{ZZ}^{-1}K_{VZ}^\top) \quad (8)$$

Let $\gamma = \{\mathbf{l}, \mathbf{Z}\}$ be the model parameters. We seek to learn the parameters by optimizing the marginal likelihood $p(\mathbf{y})$. By assuming a variational posterior over the joint signals $q(\mathbf{f}, \mathbf{u}) = q(\mathbf{u})p(\mathbf{f}|\mathbf{u})$ and following the derivation in [29], we obtain the empirical lower bound (ELBO) as in Eq. (9).

$$\log p(\mathbf{y}) \geq \mathbb{E}_{q(\mathbf{u}, \mathbf{f})} [\log p(\mathbf{y}|\mathbf{f}) - KL[q(\mathbf{u})||p(\mathbf{u})]] \quad (9)$$

Following the mean-field assumption, we specify the variational signal distribution by $q_\theta(\mathbf{u}) = \mathcal{N}(\mathbf{Z}\alpha, \mathbf{I}\sigma^2)$, where $\theta = \{\alpha, \sigma^2\}$. Optimizing the ELBO in Eq. (9) will require using Monte Carlo estimation on the first term $\mathbb{E}_{q(\mathbf{u}, \mathbf{f})} [\log p(\mathbf{y}|\mathbf{f})]$. In practice, we could first draw samples $\mathbf{u} \sim q_\theta(\mathbf{u})$ and then draw $\mathbf{f}$ from $(\mathbf{f}|\mathbf{u}) \sim p(\mathbf{f}|\mathbf{u})$ (Eq. 8). With multiple samples of $\mathbf{f} = \{\mathbf{f}^{(1)}, \dots, \mathbf{f}^{(B)}\}$, the first term in Eq. (9) can be estimated using Eq. (10).

$$\mathbb{E}_{q(\mathbf{u}, \mathbf{f})} [\log p(\mathbf{y}|\mathbf{f})] \approx \frac{1}{B} \sum_{j=1}^B \log p(\mathbf{y}|\mathbf{f}^{(j)}) \quad (10)$$

Since the KL divergence term in Eq.(9) is defined on two Gaussian distributions, the quantity can be computed analytically. With the re-parameterization trick for sampling, we are able to *end-to-end* train all the parameters γ, θ along with the parameters associated to the deep kernel.

Prediction and Uncertainty Estimation. Given the latent vectors $\mathbf{V}_*$ for the test data, the goal of prediction is to find the predictive distribution $p(\mathbf{y}_*)$, as given by Eq. (11).

$$\begin{aligned} p(\mathbf{y}_*) &= \int p(\mathbf{y}_*, \mathbf{f}_*, \mathbf{u}) d\mathbf{f}_* d\mathbf{u} \\ &\approx \mathbb{E}_{q(\mathbf{u}, \mathbf{f}_*)} [p(\mathbf{y}_*|\mathbf{f}_*)] \end{aligned} \quad (11)$$

The predictive distribution in Eq. (11) is also intractable, we again use Monte Carlo sampling to estimate the predictive mean and variance on each input latent vector. The procedure is equivalent to following Eq. (8) and (10) except that we substitute the latent vectors $\mathbf{V}$ with the $\mathbf{V}_*$.

With the prediction distribution $p(\mathbf{y}_*)$, we derive the prediction label and the corresponding uncertainty score as given by Eq. (12).

$$\text{Prediction: } \mathbb{E}_{p(\mathbf{y}_*)} [\mathbf{y}_*]; \quad \text{Uncertainty: } \text{Var}_{p(\mathbf{y}_*)} [\mathbf{y}_*] \quad (12)$$

With the pre-training initialization of the embedding networks, we jointly optimize all the hyperparameters, including deep kernel weights and the variational parameters, by backpropagation using Adam optimizer. The pseudocode of UNITE is provided in Algorithm 1.

4 EXPERIMENT

4.1 Experimental Setup

Datasets. (a) Patient EHR, demographics: We extract two patient datasets from a large longitudinal claims database¹, which include patients' EHR records as well as patients' demographics information including age, gender, and their geological locations (zip code), ranging from 2011 to 2020. The data statistics can be found in Table 2. Specifically, we conduct classification tasks for the following two diseases, i.e. Nonalcoholic fatty liver disease (NASH)

¹Dataset can be accessed upon request.

Algorithm 1: The UNITE Method

Input: Training set: $E := \{X_i, y_i\}_{i=1}^P$, $D := \{S_i\}_{i=1}^P$,
 $G := \{G_k\}_{k=1}^L, \{\gamma, \theta, w\}$

Deep kernel pre-training initialization

for $i \leftarrow 1$ **to** # pre-train epoch **do**

Sample a batch set of patients $\mathcal{B} \subset \{E, D, G\}$

Optimize the deep kernel parameters w along with a fully connected layer classifier by minimizing the cross entropy loss

end

Training with stochastic variational inference

while the *ELBO* not converged **do**

Sample a batch set of patients $\mathcal{B} \subset \{E, D, G\}$

Draw B samples of u from $q(u)$

for each patient $i \in \mathcal{B}$ **do**

Draw B samples of f_i from $p(f|u)$

Estimate $ELBO_i$ using Eq. (9) with the Monte Carlo samples (u, f_i)

end

$ELBO = \sum_{i \in \mathcal{B}} ELBO_i$

Perform stochastic gradient ascent to update $\{\gamma, \theta, w\}$ based on $ELBO$; // End-to-end training while fine-tuning w .

end

Output: Model parameters $\{\gamma^*, \theta^*, w^*\}$

and Alzheimer’s disease (AD). NASH is a liver disease that occurs in people who drink little or no alcohol. AD is a progressive disease that slowly destroys memory and thinking skills. (b) Environmental and other location-based health factors²: There is growing evidence that environmental factors and other location-based factors become important risk factors to the onset and progression of AD [15] and NASH [18]. We collect the following location-based statistics, including infant mortality rate, heart disease death rate, stroke death rate, suicide death rate, homicide death rate, drug poisoning death rate, HIV diagnosis rate, Hepatitis B cases, Hepatitis C cases, CLABSI-Standardized infection ratio, adult obesity, youth obesity, adult smoking, youth smoking, adult physical activity, youth physical activity, adult nutrition, youth nutrition, colorectal cancer screening rate, influenza vaccination coverage, and child vaccination coverage.

We join patient data with the environmental and location-based features using patients’ zip codes. Patients with the same zip code share the same location-based features G . We merge the demographics D based on the unique patient ID.

Baselines and Variants We compare with the following methods. For fair comparison, we use all datasets $\{E, S, G\}$ for all the models. For baseline methods, we concatenate the additional features (demographics and location-based features) with the EHR sequence embeddings to further perform the classification. For our proposed UNITE model, **we provide the code in the supplementary material**. For baseline models, we mainly collect the model code from the authors’ website. We defer the **code source, implementation details** and **model setup** to the appendix.

²<https://sorttablestats.cdc.gov/Index.html#/>

Table 2: Data statistics of patient EHR.

Dataset	NASH	Alzheimer
Positive	9640	29031
Negative	41777	43424
Ave. #visit	10.34	6.59
Max. #visit	105	99
Min. #visit	2	2
Ave. #medical codes	782.17	3546.03

(a) Deep Learning Models.

- LSTM [22]: the first study of applying LSTM in learning to diagnose.
- RETAIN [2]: a two-level attention-based neural model for disease risk detection.
- Dipole [24] employs a bidirectional RNN to embed the EHR data and introduces the attention mechanism to measure the inter-visit relationships.
- MiME [3] leverages the internal multilevel structure of EHR for improving risk prediction.

(b) Deep Learning Models with Uncertainty Estimations.

- MC-Dropout [11] uses dropout NNs to approximate Bayesian inference in GP. Since MC-Dropout is flexible and can be combined with any deep learning models, we build it upon the best deep learning baseline model Dipole.
- DeepEnsemble [19] models the predictive distribution with uncertainty estimation using a proper scoring rule (negative log likelihood) as training criteria and training ensembles of NNs. We build it upon the best deep learning baseline Dipole.
- DME-GP [5] leverages a deep mean function learning global effect and a deep kernel function learning the individual trend. We use RNN embedding for the best performance on risk prediction.
- INPREM [32] captures predictive uncertainty leveraging Monte Carlo sampling and designs an efficient embedding network using attention-based encoding with a linear model for interpretability.

(c) Variants of UNITE. In addition, we consider two variants of the proposed model. (1) $UNITE_d$: with a fully-connected layer as the classifier, and (2) $UNITE_b$: dropout with Monte Carlo sampling for uncertainty estimations.

Metrics Since the datasets in the experiment are imbalanced, we choose metrics that are fit for measuring model performance under data imbalance setting. (1) F1 Score; and (2) Area Under the Precision-Recall Curve (PR-AUC): popular measures of a model’s accuracy based on precision and recall, better for imbalanced setting; (3) Cohen’s Kappa (κ): $\kappa = (p_o - p_e) / (1 - p_e)$, where p_o is the observed agreement (i.e., accuracy), and p_e is the expected agreement, which is the probability of randomly seeing each category.

Table 3: Exp 1. Performance Comparison. All model performance are based on features from all modalities. UNITE has the best performance thanks to its efficient data fusion module that captures individual-level features, and the uncertainty modeling that learns population-level correlation.

	Nonalcoholic fatty liver disease (NASH)			Alzheimer’s disease (AD)		
	F1 Score	Cohen’s Kappa	PR AUC	F1 Score	Cohen’s Kappa	PR AUC
LSTM	0.309 ± 0.031	0.209 ± 0.069	0.358 ± 0.095	0.678 ± 0.040	0.469 ± 0.032	0.798 ± 0.015
MiME	0.458 ± 0.050	0.335 ± 0.068	0.483 ± 0.047	0.760 ± 0.030	0.595 ± 0.035	0.802 ± 0.013
Dipole	0.460 ± 0.021	0.325 ± 0.046	0.501 ± 0.040	0.770 ± 0.012	0.615 ± 0.014	0.827 ± 0.001
RETAIN	0.358 ± 0.032	0.226 ± 0.025	0.336 ± 0.032	0.774 ± 0.014	0.616 ± 0.009	0.822 ± 0.006
DeepEnsemble	0.505 ± 0.018	0.410 ± 0.006	0.561 ± 0.015	0.746 ± 0.069	0.884 ± 0.067	0.824 ± 0.015
MC-Dropout	0.490 ± 0.035	0.397 ± 0.019	0.546 ± 0.018	0.774 ± 0.027	0.891 ± 0.004	0.829 ± 0.010
DME-GP	0.472 ± 0.018	0.368 ± 0.021	0.452 ± 0.031	0.792 ± 0.018	0.882 ± 0.008	0.780 ± 0.013
INPREM	0.510 ± 0.033	0.363 ± 0.065	0.569 ± 0.014	0.824 ± 0.003	0.685 ± 0.004	0.843 ± 0.029
UNITE _d	0.494 ± 0.002	0.399 ± 0.009	0.517 ± 0.029	0.834 ± 0.002	0.911 ± 0.010	0.846 ± 0.017
UNITE _b	0.547 ± 0.009	0.462 ± 0.010	0.608 ± 0.004	0.841 ± 0.002	0.924 ± 0.010	0.869 ± 0.026
UNITE	0.572 ± 0.007	0.486 ± 0.009	0.609 ± 0.005	0.841 ± 0.004	0.927 ± 0.005	0.856 ± 0.020

4.2 Results

Exp 1. Performance Comparison

We experimentally evaluate UNITE against leading deep learning based risk prediction models, as well as the state-of-the-art deep learning models with uncertainty estimation on two datasets: NASH and Alzheimer’s disease (AD). NASH is often difficult to diagnose without biopsy even for experienced clinicians because many NASH patients are asymptomatic [26]. We choose the two disease detection tasks with different difficulty levels to better understand the benefit of UNITE compared with other methods. We report the results in Table 3. Results are averaged over 3 random runs. The best results are presented in bold figures.

On both datasets, UNITE significantly outperforms the best baseline method by up to 19% in Cohen’s Kappa. The performance gains of UNITE are attributed to the well-designed embedding structure and the efficient data fusion module to model the individual-level features, as well as the uncertainty modeling provided by GP, which enables the model to learn population-level correlation.

Overall, deep learning models with uncertainty estimations perform better than the models without uncertainty modeling since Bayesian modeling can better capture individual patterns and thus provide more accurate predictions. Compared with existing uncertainty estimation models, UNITE further outperforms all the baselines. With the benefit from a well-designed embedding structure in our deep kernel function, GP aims to give predictions and uncertainties by learning the global correlation among all the patients, while other uncertainty modeling methods do not capture the population-level relationships. Although DME-GP also uses GP, it does not learn the population level uncertainty but models the global trend with a multi-perceptron neural network.

Between the two datasets, UNITE provides more performance improvement compared with baselines on the NASH dataset, which shows UNITE benefits more for diseases that are highly uncertain

since NASH is difficult to diagnose without biopsy. With the predictive variances, UNITE and other Bayesian models are often more robust in terms of model mis-specification and out-of-distribution examples.

In addition, we can observe that even the variant UNITE_d (i.e., it replaces the variational inference module with a fully connected classifier) can achieve comparable performance to baselines on the NASH dataset and even better on the AD dataset. In other words, without employing GP, only using the multi-sourced data is effective for risk prediction. To further improve the performance and provide the utility of the designed model, we investigate the uncertainty-based variant UNITE_b, which estimates uncertainty with Monte Carlo sampling to approximate Bayesian NNs, performs better than UNITE_d. This result demonstrates considering uncertainty in risk prediction is necessary. However, UNITE achieves the best overall performance among all the baselines and its two variants, which conforms the reasonableness of the designed model and the effectiveness of GP for risk prediction.

Exp 2. Leverage Uncertainty to Improve Prediction

In clinical decision making where overconfident incorrect predictions can be harmful or offensive, it would be better to set the highly uncertain subjects aside for human experts to evaluate. Also, it is always better to continue improving the prediction score to reach some thresholds that are reliable enough for decision making.

To investigate whether leveraging UNITE and the learned uncertainty score could potentially identify truly uncertain patients that are easily misdiagnosed and thus improve the prediction, we perform the analysis below. We remove the patients with high uncertainty and check the improvement of the performance. We remove the top 20%, 50%, and 80% of samples based on their uncertainty scores respectively for comparison. The results are listed in

Table 4: Exp 2. Leverage Uncertainty to Improve Prediction. We remove the top 20%, 50% and 80% of patients based on their uncertainty scores, respectively. The model achieves higher score gradually, especially for NASH that is highly uncertain and thus difficult to detect.

NASH	F1 Score	Cohen’s Kappa	PR AUC
0%	0.572 $\pm$ 0.007	0.486 $\pm$ 0.009	0.609 $\pm$ 0.005
20%	0.592 $\pm$ 0.028	0.501 $\pm$ 0.037	0.587 $\pm$ 0.007
50%	0.608 $\pm$ 0.059	0.522 $\pm$ 0.076	0.602 $\pm$ 0.020
80%	0.621 $\pm$ 0.080	0.537 $\pm$ 0.100	0.610 $\pm$ 0.012
AD	F1 Score	Cohen’s Kappa	PR AUC
0%	0.840 $\pm$ 0.004	0.725 $\pm$ 0.005	0.856 $\pm$ 0.020
20%	0.858 $\pm$ 0.032	0.754 $\pm$ 0.060	0.890 $\pm$ 0.010
50%	0.868 $\pm$ 0.051	0.763 $\pm$ 0.079	0.894 $\pm$ 0.020
80%	0.880 $\pm$ 0.069	0.763 $\pm$ 0.074	0.905 $\pm$ 0.037

Table 4. As expected, for NASH that is highly uncertain for diagnosis, leveraging uncertainty score and ruling out highly uncertain patients could significantly improve the all accuracy measures for the rest of the cohort. For AD which has more distinguishing disease patterns, the performance could still be improved.

Exp 3. Effect of Multiple Data Sources

We conduct an ablation study to understand the contribution of each data sources in UNITE. We remove/change the data modalities as below and compare the accuracy scores. The parameters in all the variants are determined with cross-validation, and the best performance is plotted in Fig. 2. We compare UNITE with the reduced models below.

- UNITE-G-D: without using location-based data G and demographics D , only EHR sequences E are used for training and inference.
- UNITE-G: without using location-based features, trained on patient EHR data E and demographics D .

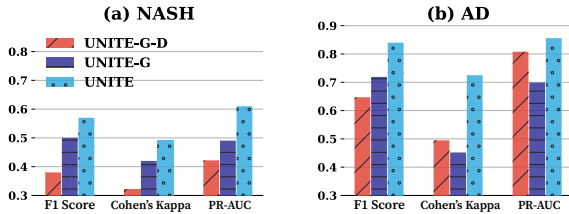


Figure 2: Exp 3. Effect of Multiple Data Sources.

From Fig. 2, it can be observed when we solely use the patient EHR data for disease risk prediction, the performance drops largely especially for NASH which is hard to diagnose. It suggests the necessity of using all data including patient demographics and location-based features.

Exp 4. Clustering Effect of UNITE

To investigate their effectiveness of learned covariance matrix, we compare two GP models, UNITE and DME-GP. We randomly sample an unseen dataset including 200 patients (100 positive and 100 negative) and expect that the models could perfectly cluster the population into 2 groups. Using spectral biclustering [17], we visualize the covariance matrices as shown in 4. UNITE successfully clusters the population into 2 groups, while DME-GP fails learn any meaningful correlations among patients. We defer the experiments details in Appendix.

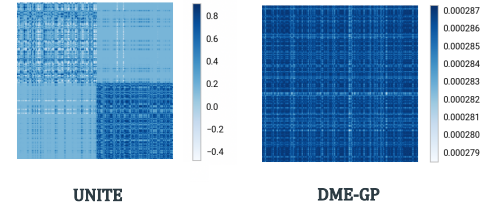


Figure 3: Exp 4. Clustering Effect of UNITE. UNITE successfully clusters the patients.

5 CONCLUSION

In this paper, we propose UNcertaINTy-based hEalth risk prediction (UNITE) model, that it built upon an adaptive multimodal deep kernel and a stochastic variational inference module, to provide accurate disease risk prediction and uncertainty estimation leveraging multi-sourced health data including environmental factors. Experiments on real-world datasets demonstrate the strong performance of UNITE. The UNITE method can also be extended to other application domains where overconfident incorrect predictions can be harmful or offensive, such as fraud detection.

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6 APPENDIX

6.1 Implementation Details

For each task (AD and NASH), we randomly split the dataset into training, validation, and testing sets in a 64:16:20 ratio. The medical code embedding is set as 128 for all models. The vocabulary size is counted separately, 26706 for NASH and 31467 for AD. The sequence length is chosen as 200 according to the data statistics. For training all approaches, we use Adam [16] with the batch size of 256 and the learning rate of $1e-3$. We use the cross entropy loss. All models are trained on Ubuntu 16.04 with 128GB memory and a Nvidia Tesla P100 GPU.

UNITE Setup. For the patient EHR record embedding, we use 2 blocks of transformers, each block applies a multi-head self attention layer, a normalization layer, a feed forward layer and another normalization layer. For demographics and location-based feature embeddings, we both set 2 fully-connected layers to learn the latent representations. Other hyperparameters are shown in Table 5.

Baseline Setup. We implement RETAIN³, Dipole⁴, MiME⁵, Deep-Ensemble⁶, MC-Dropout⁷, and DME-GP⁸ based on the available open-sourced code repositories. For INPREM, we acquired the code from the authors. We implement LSTM ourselves. For each of the baeline models, we grid search for the best hyperparameters. The training process is done with 30 epochs and the Adam optimizer. The hyperparameters are shown in Table 5.

6.2 Metrics

(1) **F1 Score**: $= 2 \cdot (\text{Prec} \cdot \text{Rec}) / (\text{Prec} + \text{Rec})$, where Prec is precision and Rec is recall.

(2) **Cohen’s Kappa (κ)**: can be used to measure the performance of the different classifiers for the same dataset, especially useful for imbalanced dataset.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (13)$$

p_o is the observed agreement, which is identical to accuracy. p_e is the expected agreement, which is the probability of random seeing each category.

(3) **Area Under the Precision-Recall Curve (PR-AUC)**:

$$\text{PR-AUC} = \sum_{k=1}^n \text{Prec}(k) \Delta \text{Rec}(k),$$

where k is the k -th precision and recall operating point ($\text{Prec}(k)$, $\text{Rec}(k)$).

6.3 Related Work on Combining GP and deep learning models

There has been a growing interest in combining deep architectures with GP to benefit from both methods. One way is to transform the inputs of the base kernels with a deep architecture. [13] propose to use a deep belief networks to learn a good covariance kernel. [14] applied deep neural networks as the feature-mapping function of GPs. More generally, [29] use the similar idea by designed the

³<https://github.com/easyfan327/Pytorch-RETAIN>

⁴<https://github.com/yzhao062/PyHealth>

⁵<https://github.com/mp2893/mime>

⁶<https://github.com/Kyushik/Predictive-Uncertainty-Estimation-using-Deep-Ensemble>

⁷<https://github.com/cpark321/uncertainty-deep-learning>

⁸<https://github.com/jik0730/Deep-Mixed-Effect-Model-using-Gaussian-Processes>

Table 5: Summary of all models hyperparameters

Hyperparameters	Value
For all models	
Batch size	256
Learning rate	$1e-3$
Input dim for EHR sequence	200
Input dim for demographics	2
Input dim for location-based features	34
Hidden dim for EHR sequence	16
Hidden dim for demographics	2
Hidden dim for location-based features	2
UNITE	
# Inducing points	200
Hidden dim for fused vector	16
Transformer depth	2
Transformer # head	8
DeepEnsemble	
# ensemble models	10
Base model	Dipole
MC-Dropout	
Dropout rate	0.1
Base model	Dipole
DME-GP	
Mean function	MLP
Embedding function	LSTM

kernels parameterized by neural networks to augment the model with the expressiveness power from deep architectures. Another line of works [6, 27] proposed a stacked GP model which has multiple layers of Gaussian Processes. In this work, we augment the first framework by learning a deep multimodal kernel function parameterized by well-designed deep neural networks.

6.4 Visualizing the Biclustering of Covariance Matrix

We construct a smaller dataset which includes a random sample of 200 patients (100 positive and 100 negative). We learn the covariance matrix $\Sigma = K_{VV}$ as specified in Eq. (5). Specifically, element $\sigma_{i,j}$ in the i -th row and j -th column represents the correlation between sample i and sample j . The diagonal of σ consists of the variance σ_i^2 of the i -th sample. To visualize the effect of learned correlations, we use spectral biclustering algorithm [17] to cluster the rows and columns of Σ . The learned covariance matrix are visualized in the top row of Figure 4. After biclustering, patient clusters are shown in the bottom row of Fig. 4. For each dataset, the covariance matrix successfully clusters the whole population into 2 groups with different strength of correlation, aligned with the groundtruth where each group contains 100 samples. This proves that UNITE captures the population-level correlations and can potentially help find similar patients to support clinical decision making. Interestingly, in the

NASH case as shown in Fig. 4 (a), both the negative and positive samples show strong correlations (dark color), while for AD as shown in Fig. 4 (b), only the positive group of patients show strong correlations.

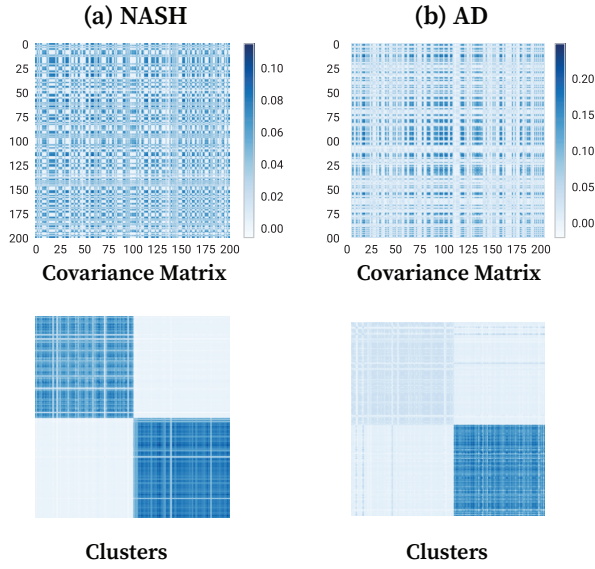


Figure 4: Exp 4. Leverage Covariance Matrix for Similar Patient Retrieval. The covariance learned by UNITE successfully clusters the patients, aligned with the ground-truth.